Summary of Resonance Field Theory and Empirical Results

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1 Introduction and Objectives

Resonance Field Theory (RFT) is an axiomatic framework that describes oscillatory coupling as a universal organizing principle of physical and systemic phenomena. The goal is a parametric extension of the Planck relation that uniformly captures resonance phenomena across scales and domains – from particle physics through cosmology to financial markets and nuclear technology.
This document summarizes the theoretical foundations, the methodology, and the empirical results. The complete manuscript in IOP format is available as `rft_manuscript_en_iop.tex`. All derivations, source codes, and data are openly accessible at `https://github.com/DominicReneSchu/RFT`.
Philosophical and metaphysical extensions of the RFT are separately documented as interpretative extensions (`en/facts/docs/explanations/the_universe_as_a_resonance_bubble.md`) and explicitly marked as speculative perspectives.

2 Theoretical Background

The RFT formulates seven hierarchically structured axioms (A1–A7), from which the central *Resonance Field Equation* follows:

$$E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f \tag{1}$$

Here:

- π is the mathematical constant as a measure of cyclic structure,
- $\hbar = h/(2\pi)$ is the reduced Planck action quantum[1],
- f is the resonance frequency of the system,
- $\varepsilon(\Delta\varphi) \in [0, 1]$ is the phase-dependent **coupling operator**, which describes the strength of the resonance coupling between two systems as a function of their phase difference $\Delta\varphi$.

2.1 π as a Geometric Fundamental Constant of Phase Space

π appears in the Resonance Field Equation not as an arbitrary numerical prefactor. It is the **natural measure of a half-oscillation**: the phase interval $[0, \pi]$ corresponds to the transition from full phase coherence ($\Delta\varphi = 0$, $\varepsilon = 1$) to complete destructive interference ($\Delta\varphi = \pi$, $\varepsilon = 0$).
Status (RT-01, RT-01b): The factor π is not a free parameter, but the **geometrically derived saddle-point contribution** of the stationary action integral $S[\psi, \Delta\varphi]$. The limit $E = \pi\varepsilon\hbar f \to E = hf_{\mathrm{Hz}}$ is formally closed. Numerical confirmation (RT-01b): convergence to π to machine precision, $|c_3 + c_4| \approx 5.5 \times 10^{-11}$.

2.2 Conceptual Distinctions

- **Coupling operator** $\varepsilon(\Delta\varphi)$: Dimensionless parameter in the basic formula. Representation-theoretically uniquely determined as the only function belonging exclusively to the k=1 representation of $\mathrm{U}(1) \subset G_{\mathrm{sync}}$ (RT-02): $\varepsilon = \cos^2(\Delta\varphi/2)$. Limit values: $\varepsilon = 0$ (anti-resonance), $\varepsilon = 1$ (complete phase coherence).
- **Coupling efficiency** $\eta(\Delta\varphi)$: Measurable quantity in the FLRW simulations. Extracted from the time-averaged cross-term of two coupled scalar fields. Analytical expectation: $\eta_{\mathrm{theo}} = \cos^2(\Delta\varphi/2)$.
- **Identity** $\varepsilon = \eta$: In the resonance reactor context it was shown that $\varepsilon(\Delta\varphi) = \eta(\Delta\varphi)$ holds exactly. This identity eliminates the coupling strength parameter κ completely: $\kappa = 1$ (no free parameter).

[1] The RFT uses $\hbar$ (reduced Planck constant) throughout, which is the canonical choice for angular-frequency formulations. The relation to the ordinary Planck constant is $\hbar = h/(2\pi)$.

2.3 Planck Special Case

For $\varepsilon = 1/(2\pi)$ it follows that $E = \frac{1}{2}\hbar f$, consistent with the ground-state energy of the harmonic oscillator.

2.4 The Seven Axioms (Axiom Status Aug. 2026)

A1 **Universal Oscillation:** Every physical entity can be described by an oscillation function. *Status: Postulate.*

A2 **Superposition and Interference:** Oscillations superpose linearly. *Status: Postulate.*

A3 **Resonance Condition:** Coupling occurs when frequencies stand in rational ratios ($f_1/f_2 = n/m$). *Status: Corollary from A7 (RT-02, RT-35); not an independent axiom.*

A4 **Coupling Energy:** $E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f$. *Status: π geometrically derived (RT-01, RT-01b); ε representation-theoretically unique (RT-02).*

A5 **Energy Direction:** Energy is a vectorial quantity with magnitude and direction in the resonance field. *Status: Irreducible postulate (RT-36); not derivable from G_{sync}.*

A6 **Information Flow:** Information propagates exclusively along coherent resonance paths. *Status: Postulate.*

A7 **Invariance under G_{sync}:** Resonance structures are preserved under variation of the observer's perspective and system parameters; $G_{\text{sync}} \cong \mathbb{R}^{+}_{\times} \times \mathrm{U}(1) \times \mathrm{Aff}^{+}(\mathbb{R})$. *Status: Algebraically proved (stationary), RT-02.*

3 Methodology and Software Architecture

The empirical validation is carried out by four independent simulation pipelines:

3.1 Monte Carlo Resonance Analysis

- 50 000 pseudo-experiments on CMS open data (99 915 dielectron events)
- KDE background model with narrow signal exclusion
- Binomial test with Bonferroni correction over 68 window widths
- Bootstrap confidence intervals (10 000 repetitions)
- Systematics checks over three KDE bandwidths and ten independent seeds
- **Total scope:** 1 500 000 **simulations**

3.2 Coupled FLRW Simulations

- 1 530 individual simulations (51 H_0 points × 30 phase values)
- Klein-Gordon equation in expanding spacetime
- DOP853 integrator, $t = 0$–120, 12 000 evaluation points
- Jackknife error on d_η, bootstrap error on slope

3.3 Resonance Reactor Simulation

- GDR data from literature: Berman & Fultz (1975), RIPL-3, Ishkhanov & Kapitonov (2021)
- Resonance frequencies from RFT basic formula: $f = E_{\mathrm{GDR}}/(\pi \cdot \hbar)$
- Coupling strength $\kappa = 1$ from $\varepsilon = \eta$ identity
- Energy balance with Q-factor analysis

3.4 ResoTrade (Application Concept)

- Resonance field theory trading system for the BTC market – demonstrates applicability of RFT axioms in a real market environment
- AC/DC decomposition (A1), superposition of three modes (A2), frequency ratio 24h:168h = 1:7 (A3), balance regulator and downtrend pause gate (A4), energy direction vector (A5), resonance gate (A6)
- Primary axiom evidence comes from RFT-internal simulations (FLRW, Monte Carlo, double pendulum, resonance reactor)

4 Results

4.1 Monte Carlo on CMS Dielectron Data

Five resonances with empirical p-value = 0 detected (1 500 000 total simulations):

Resonance	M_0 (GeV)	**Hits**	p_{corr}	**emp.** p
$\phi(1020)$	1,020	19 869	$1{,}07 \times 10^{-54}$	0
J/ψ	3,100	3 256	$1{,}02 \times 10^{-121}$	0
Υ(1S)	9,460	4 186	$3{,}01 \times 10^{-201}$	0
Υ(2S)	10,020	4 625	$5{,}16 \times 10^{-188}$	0
Z-boson	91,200	52	$< 10^{-300}$	0

Bandwidth-independent (0,3, 0,5, 0,7 GeV) and seed-independent (10 seeds, all emp. $p = 0{,}000 \pm 0{,}000$). Confirms Axioms 3 and 7.

4.2 Coupled FLRW Simulations

Measured quantity	**Value**
Individual simulations	1 530
Slope $\mathrm{d}d_\eta/\mathrm{d}H_0$	$(0{,}00113 \pm 0{,}00017)$ (km/s/Mpc)$^{-1}$
Δd_η (SH0ES − Planck)	$0{,}0063 \pm 0{,}0010$ ($> 6\sigma$)
$\Delta\chi^2$ vs. Planck-2018-CMB	+16,0

Confirms Axioms 1, 3, 4, 5, 7.

4.3 Resonance Reactor: Resonant Transmutation

4.3.1 Results ($\Phi_\gamma = 10^{12}\ \gamma/(\mathrm{cm}^2{\cdot}\mathrm{s})$, $\Delta\varphi = 0$, $\kappa = 1$).

Isotope	$t_{1/2}$	$\lambda_{\mathrm{eff}}/\lambda_0$	Q_{fiss}	**Interpretation**
U-235	$7{,}04 \times 10^8$ y	7 872	0,97	Resonance dominates
Pu-239	24 110 y	1,3	1,00	Break-even
Am-241	432 y	1,005	–	Marginal

Falsifiable prediction: $\sigma_{\mathrm{coh}} > \sigma_{\mathrm{incoh}}$. Confirms Axioms 1, 3, 4.

4.4 ResoTrade: Application Concept on the BTC Market

ResoTrade is a resonance field theory trading system that applies all seven axioms to the concrete problem of algorithmic trading. It demonstrates the applicability of the RFT axioms in a real, chaotic system – but does not serve as primary axiom evidence, since the underlying implementation is not publicly verifiable.

4.4.1 Dynamic simulations.

The RFT-internal simulations confirm the axioms independently and verifiably:

Axiom	Simulation	Result
A1	FLRW (1 530 runs)	$\eta \approx \cos^2(\Delta\phi/2)$, $\Delta d_\eta > 6\sigma$
A2	Coupled oscillators	Multi-frequency superposition confirmed
A3	Monte Carlo (1 500 000 sim.)	5 resonances, emp. $p = 0$
A4	Schrödinger sim. (5 stages)	Fidelity = 1.0 (all 4 scenarios), $1 - F \sim \lambda^2$
A4	Warp drive sim.	First positive-energy warp bubble, w sign change
A4	Resonance reactor	$\lambda_{\text{eff}}/\lambda_0 = 7\,872$ (U-235), $\kappa = 1$
A4	FLRW: $\varepsilon = \eta$ identity	$\kappa = 1$, no free parameter
A5	Resonance field sim.	Energy direction vector confirmed
A5	Warp drive sim.	Front/rear asymmetry (contraction vs. expansion)
A6	Resonance field sim.	Coupling efficiency and energy flow
A7	Monte Carlo	bandwidth-independent (3 KDE)

Confirms Axioms 1, 2, 3, 4, 5, 6, 7.

4.5 Dynamic Simulations

Double pendulum model and Schrödinger simulation for visualization of complex coupling states:

- Double pendulum: known motion patterns confirmed, novel resonance modes identified, $\varepsilon(\theta_2 - \theta_1) = \cos^2(\Delta\theta/2)$ dynamically confirmed (A1, A2, A4)
- Schrödinger simulation: derivation of the Schrödinger equation from Axiom 4 in 5 derivation steps; Fidelity = 1,0 (all 4 scenarios); perturbation theory $1 - F \sim \lambda^2$ confirmed; falsifiable prediction $|\Delta\langle x\rangle| \approx 2{,}0 \cdot \lambda$ μm for ^{87}Rb (A4)
- Warp drive simulation: first positive-energy warp bubble ($E^- = 0$); sign change in equation of state w via $\varepsilon(\Delta\varphi)$ phase control; $\Delta w = +0{,}057$; $\rho > 0$ everywhere (A4, A5)

5 Mathematical Proofs

Through new methods of series expansion it has been possible to:

- analytically solve previously unsolved integrals,
- establish systemic connections between frequency, coupling, and energy,
- prove the identity $\varepsilon(\Delta\varphi) = \eta(\Delta\varphi)$, which eliminates the last free parameter.

6 Outlook and Open Questions

1. **Independent determination of ε:** Ab-initio derivation from system geometry.
2. **New prediction:** $\sigma_{\text{coh}} > \sigma_{\text{incoh}}$ experimentally testable in the resonance reactor.
3. **Resonance reactor – $Q > 1$:** Break-even curve $d_{\min}(k)$, optimal target geometry.

4. **CMB with Boltzmann solver:** Full ℓ-range with CAMB/CLASS.
5. **Nonlinear saturation:** $\lambda\varepsilon^4$- and $\alpha R \varepsilon^2$-terms.
6. **Resonance Hamiltonian:** Construct and solve $\hat{H}_{\text{res}}$ for specific quantum systems (phonon coupling, spin-orbit coupling).
7. **Warp drive – 3D bubble:** Full 3D warp geometry with spherical-azimuthal symmetry; quantify energy gap at stage 5.
8. **Further domains:** Acoustics, biology, network theory.

7 Experimental Predictions

The RFT formulates two falsifiable experimental proposals that test the coupling function $\cos^2(\Delta\varphi/2)$ over 19 orders of magnitude in energy.

7.1 Experiment I: Am-241 at ELI-NP (Nuclear physics)

- **Prediction:** Signal(coherent) / Signal(incoherent) = 2,0 (RFT) vs. 1,0 (SM). No free parameters ($\kappa = 1$ from $\varepsilon = \eta$).
- **System:** 100 mg Am-241, GDR centroid 14,0 MeV
- **Facility:** ELI-NP VEGA (Măgurele, Romania), $\Phi > 10^{11}$ γ/s, $\Delta E/E < 0{,}5\%$
- **Measurement protocol:** 5 measurement points (M1–M5) with varying phase coherence. M5 = reference (incoherent). RFT predicts: $\cos^2(\Delta\varphi/2)$ pattern over M1–M5. SM predicts: all equal.
- **Significance:** $> 50\,000\,\sigma$
- **Beam time:** ~30 h, cost ~50 000 EUR

7.2 Experiment II: ^{87}Rb-BEC in harmonic trap (Quantum mechanics)

- **Prediction:** $|\Delta\langle x\rangle| = 4{,}9 \cdot \lambda \cdot \ell \approx 2{,}0 \cdot \lambda$ μm (RFT) vs. 0 (Standard QM). One free parameter (λ).
- **System:** Ultracold ^{87}Rb atoms in harmonic trap, $\omega = 2\pi \times 100$ Hz
- **SI calibration:** $\ell = 0{,}41$ μm, $\tau = 0{,}225$ ms, consistency over 6 independent relations verified ($\hbar = m \cdot \ell^2/\tau$ etc.)
- **Perturbation theory:** Numerically verified over λ-scan ($10^{-4} \ldots 5$): $1 - F \sim \lambda^{2{,}00}$, $\Delta\langle O\rangle \sim \lambda^{1{,}00}$. Rel. error $1{,}3 \times 10^{-4}$.
- **Kohn theorem:** GP interaction does not shift $\langle x\rangle$ (Kohn-protected). RFT feedback breaks Kohn condition – unique signal.
- **Sensitivity:** $\lambda \gtrsim 0{,}05$ after 100 repetitions, $\lambda \gtrsim 0{,}005$ after 10 000

7.3 Comparison of Both Experiments

	Am-241 (Nuclear physics)	**^{87}Rb-BEC (QM)**
Domain	Nuclear physics (GDR, 14 MeV)	QM (μeV)
Energy scale	10^7 eV	10^{-12} eV
Measured quantity	$\text{Signal}_{\text{coh}}$ / $\text{Signal}_{\text{inc}}$	$\Delta\langle x\rangle$
RFT prediction	2,0 (exact)	$2{,}0 \cdot \lambda$ μm
SM prediction	1,0	0
Free parameters	0	1 (λ)
Tested quantity	$\eta(\Delta\varphi) = \cos^2(\Delta\varphi/2)$	$\varepsilon(\Delta\varphi(t))$ dynamic
Test type	Yes/No (ratio)	Continuous (λ-scan)

Both experiments test the same core of the RFT (the coupling function $\cos^2(\Delta\varphi/2)$) over 19 orders of magnitude in energy. They are independently falsifiable.

Document Overview

Document	Content
`rft_manuscript_en_iop.tex`	Physics, simulations, peer review
`RFT_Summary.tex`	Technical overall overview

Appendix: Directory Structure

Monte Carlo: `en/facts/empirical/monte_carlo/`
FLRW: `en/facts/simulations/FLRW_simulations/`
Schrödinger sim.: `en/facts/simulations/schrodinger/`
Resonance reactor: `en/facts/concepts/resonance_reactor/`
Dynamic sim.: `en/facts/simulations/double_pendulum/`

Repository: `https://github.com/DominicReneSchu/RFT`